MARYAM REHAN

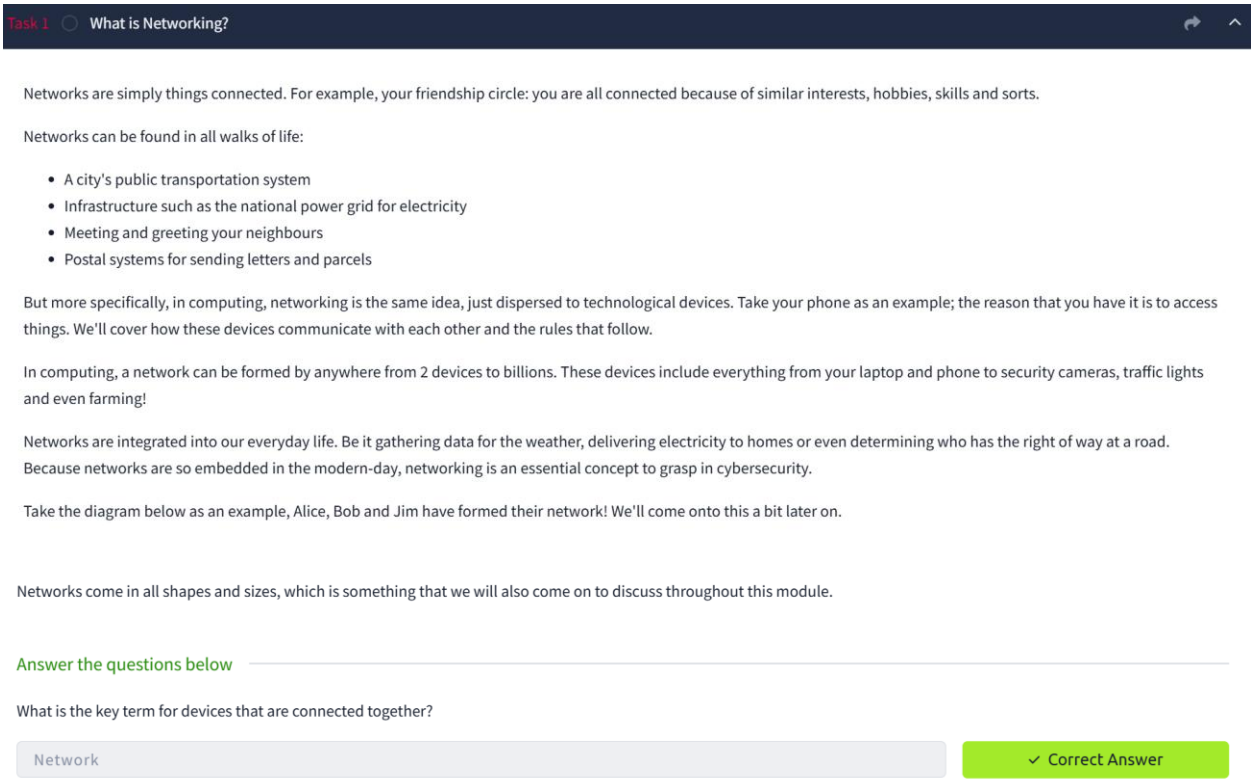
INTERNSHIP WEEK 01 TASK 02

CYBER SECURITY (SOC)

NETWORKING FOUNDATION FROM A DEFENDER'S VIEW

Internship Task: Networking Fundamentals
Platform: TryHackMe

NETWORKING CONCEPTS



1. Objective

The objective of this task is to understand the fundamentals of computer networking from a defensive cybersecurity perspective. This includes learning how devices communicate, understanding network protocols and ports, exploring the OSI model, and recognizing normal and potentially suspicious network traffic.

2. Introduction to Networking

A computer network is a group of two or more devices connected to communicate and share information or resources. These devices may include computers, laptops, smartphones, servers, printers, and security cameras.

Networking is an essential part of cybersecurity because systems communicate over networks to exchange data. Security analysts monitor this communication to identify unusual activity, unauthorized access, and potential cyber threats.

3. Practical Activity — TryHackMe

Platform: TryHackMe
Room: What is Networking?
Section completed: Task 1 — What is Networking?

Question

What is the key term for devices that are connected together?

Answer: Network

Learning Outcome

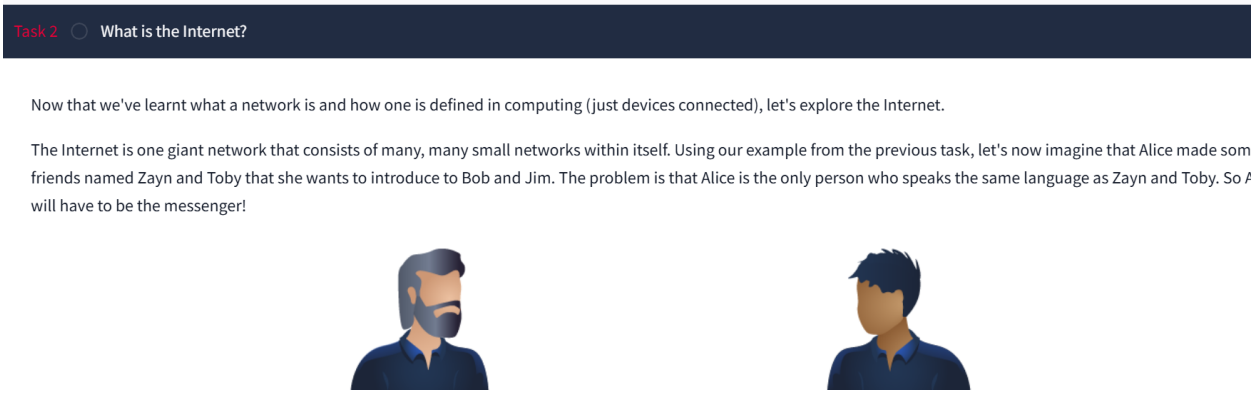
I learned that connected devices form a network and that computer networking enables devices to communicate and share information. Understanding this concept is a foundation for learning network security and Security Operations Center (SOC) monitoring.

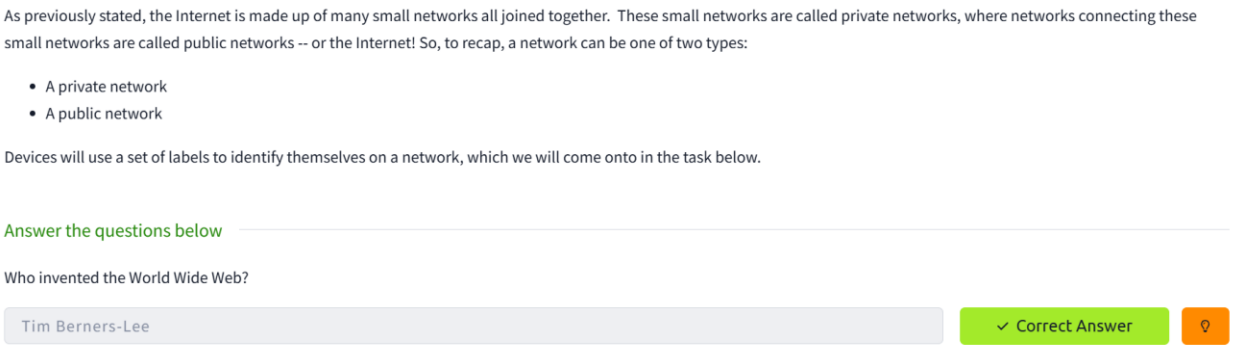
4. Evidence

Insert a screenshot of the completed Task 1 question and its accepted answer here.

5. Conclusion

This practical activity introduced me to the basic concept of computer networking. I learned that networks connect devices and enable communication, which is essential for understanding how a SOC analyst monitors network activity and investigates potential security incidents.





1.Objective

To understand the Internet and the World Wide Web (WWW), including their basic concepts and historical background.

2.Question and Answer

Question: Who invented the World Wide Web?

Answer: Tim Berners-Lee.

3.Explanation

The World Wide Web was invented by Tim Berners-Lee in 1989. It allows users to access and navigate interconnected web pages through web browsers.

ARPANET was an early packet-switching network that played an important role in the development of the modern Internet. However, ARPANET did not invent the World Wide Web.

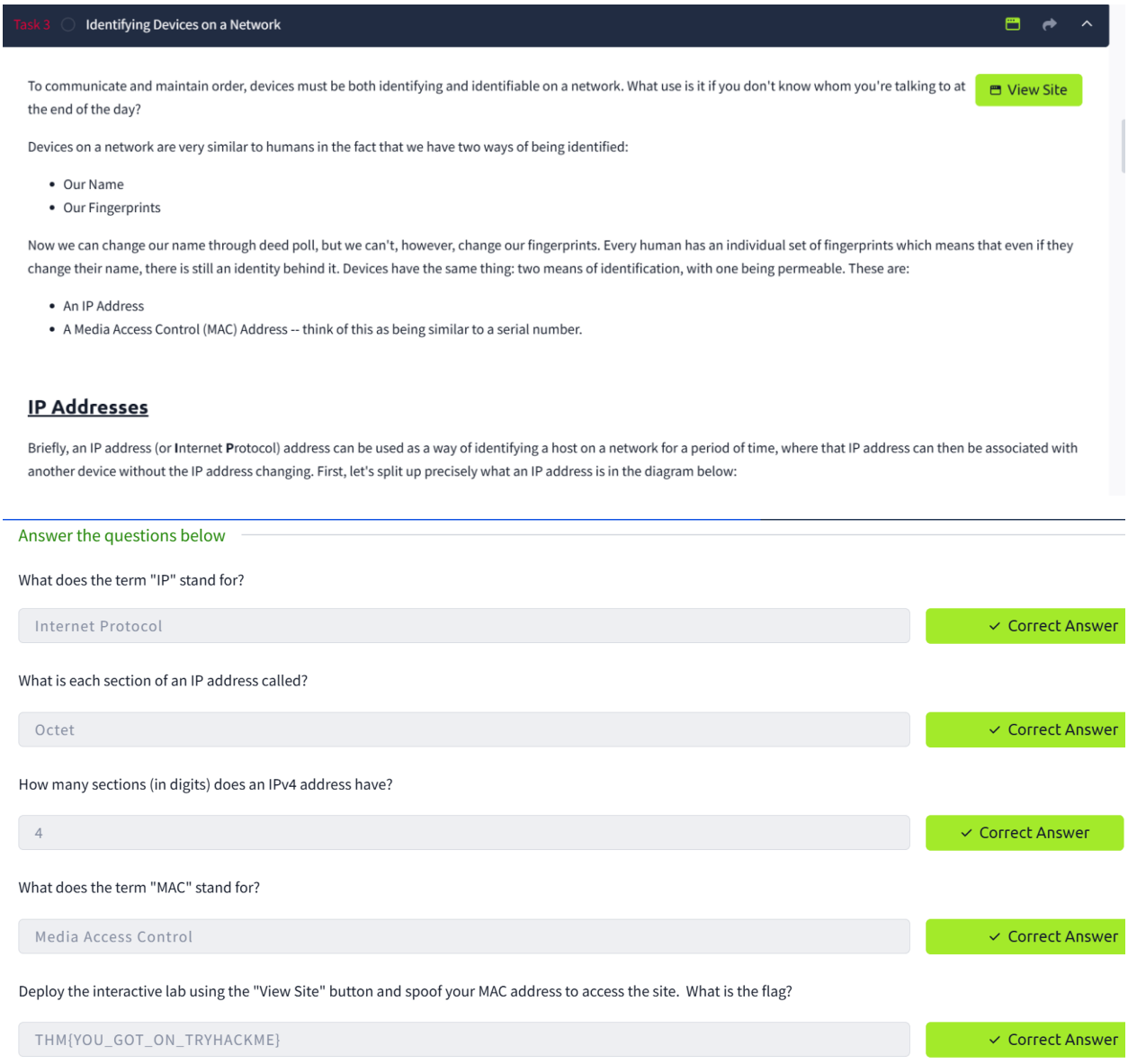
The Internet is a global network of interconnected computer networks, while the World Wide Web is a service that operates over the Internet.

4.Learning Outcome

I learned the difference between the Internet and the World Wide Web. I also learned that Tim Berners-Lee invented the World Wide Web, whereas ARPANET was an important predecessor to the modern Internet.

5.Conclusion

In conclusion, I learned that the Internet is a global network of interconnected computers and devices that communicate and share information. I also learned that the World Wide Web (WWW) was invented by Tim Berners-Lee in 1989. The Internet provides the infrastructure for services such as the Web, email, and online communication. Understanding the difference between the Internet and the WWW is an important foundation for learning networking and cybersecurity.



1.Objective

The objective of this practical activity was to understand how devices are identified on a network using IP addresses and MAC addresses, and to demonstrate how MAC address spoofing can affect network access controls.

2. Introduction

Devices communicate over networks using addressing systems that help identify them. IP addresses are used for logical addressing, while MAC addresses identify network interfaces on a local network. Understanding these addresses is important for network administration and cybersecurity monitoring.

3. IP Addresses

An Internet Protocol (IP) address is a logical address used to identify a device on a network.

There are two main versions of IP addresses:

- **IPv4:** Uses a 32-bit address represented in four octets, such as 192.168.1.77.
- **IPv6:** Uses a 128-bit address and provides a much larger address space.

IP addresses may be public or private. Public IP addresses are used for Internet-facing communication, while private IP addresses are commonly used within local networks.

4. MAC Addresses

MAC stands for Media Access Control. A MAC address is associated with a network interface and is commonly represented as six pairs of hexadecimal characters separated by colons.

Example: a4:c3:f0:85:ac:2d

MAC addresses are used for communication on a local network segment. Unlike IP addresses, which can change depending on network configuration, MAC addresses are generally assigned to network interfaces, although they can be changed or spoofed through software.

5. Practical Activity: Hotel Wi-Fi Lab

Platform: TryHackMe
Room: What is Networking?
Task: Task 3 — Identifying Devices on a Network

Procedure

1. Opened the interactive Hotel Wi-Fi lab.
2. Observed the simulated network containing Bob, Alice, a router, and a website.
3. Identified that Alice had authorized Wi-Fi access while Bob's traffic was initially blocked.
4. Changed Bob's MAC address to match Alice's authorized MAC address.
5. Requested access to the website and observed the successful result.

Result

The lab was completed successfully, and the following flag was obtained:

THM{YOU_GOT_ON_TRYHACKME}

6. Security Analysis

The practical exercise demonstrated that relying only on MAC address filtering can be insufficient for protecting network access. An attacker may spoof an authorized MAC address and impersonate another device if the network does not use stronger authentication controls.

Recommended security measures include:

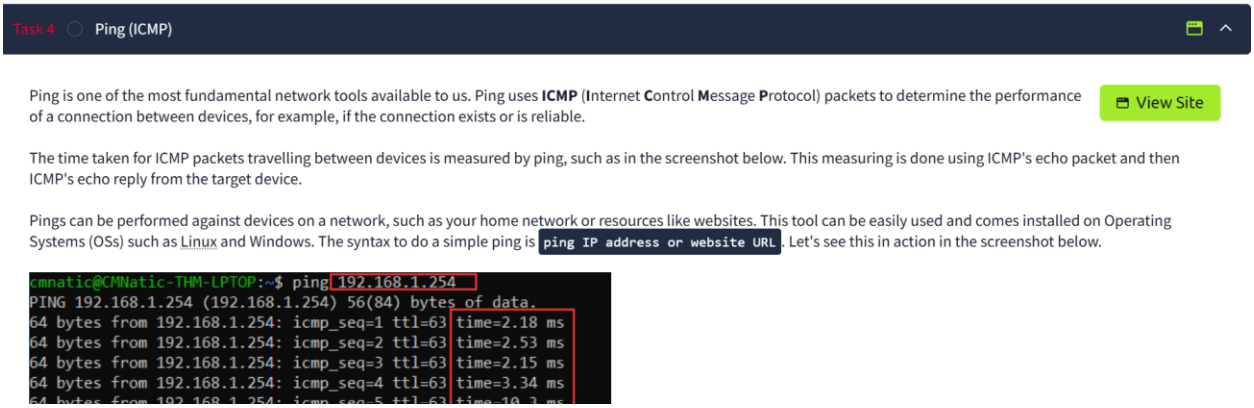
- Use strong authentication, such as WPA2-Enterprise or WPA3-Enterprise where appropriate.
- Monitor unusual device connections and duplicate MAC address activity.
- Apply network segmentation and access controls.
- Investigate suspicious connections through network monitoring and security logs.

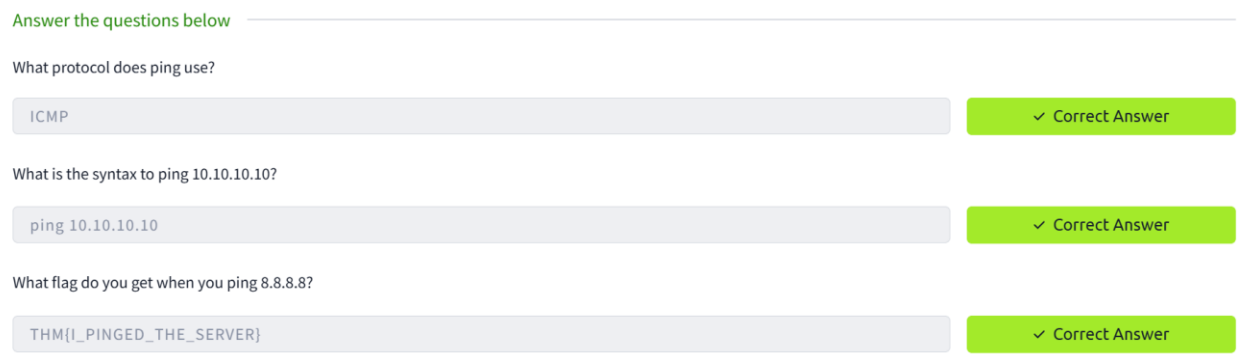
7. Learning Outcomes

Through this activity, I learned the difference between IP and MAC addresses, the structure of IPv4 addresses, and the purpose of public and private IP addressing. I also demonstrated MAC address spoofing in an authorized educational lab and learned why MAC filtering alone is not a reliable security mechanism.

8. Conclusion

This practical activity improved my understanding of network addressing and device identification. The Hotel Wi-Fi simulation demonstrated how weak MAC-based access controls can be bypassed and highlighted the importance of stronger authentication and continuous network monitoring in cybersecurity.





Objective

To understand the ICMP protocol and use the ping command to test network connectivity between devices.

Introduction

Ping is a basic network diagnostic tool used to check whether a target device is reachable over a network. It uses the Internet Control Message Protocol (ICMP) to send Echo Request packets and receive Echo Reply packets. It also measures the response time between devices.

Practical Implementation

The TryHackMe Ping (ICMP) lab was used to ping the IP address 8.8.8.8. The command executed was:

ping -c 4 8.8.8.8

The -c 4 option sends four ping requests and then stops the command.

Results and Observations

The ping test completed successfully. All four packets were transmitted and received, with zero packet loss. This indicated that the target was reachable from the lab environment.

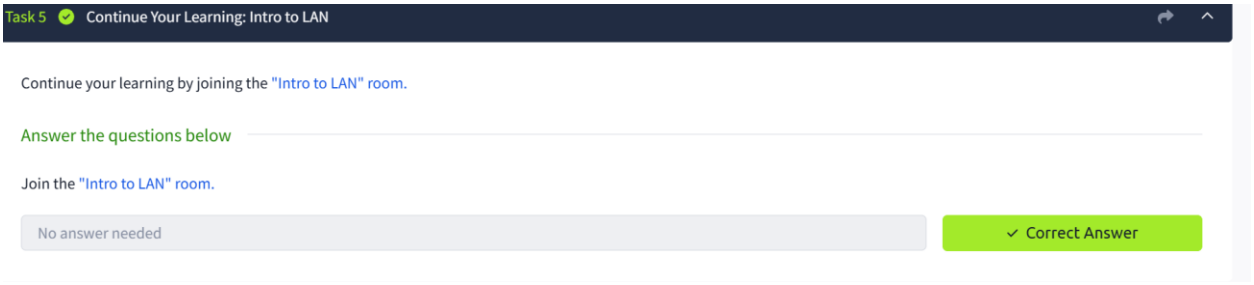
- **Protocol:** ICMP
- **Target IP:** 8.8.8.8
- **Packets transmitted:** 4
- **Packets received:** 4
- **Packet loss:** 0%
- **TryHackMe flag:** THM{I_PINGED_THE_SERVER}

Security Relevance

Ping can help network defenders troubleshoot connectivity issues and identify whether a host responds to ICMP requests. However, a successful ping does not prove that all services on a host are available or secure. Unexpected ICMP traffic may also be worth investigating as part of network monitoring.

Conclusion

This practical helped me understand how the ICMP protocol works and how the ping command is used to test network connectivity. I successfully pinged 8.8.8.8 in the TryHackMe lab and received all four responses without packet loss. I also learned that ping is useful for basic troubleshooting, but its results should be considered alongside other network monitoring tools.



Objective

To understand the fundamentals of a Local Area Network (LAN) and how devices communicate within a local network.

Introduction

A Local Area Network (LAN) connects devices within a limited geographical area, such as a home, office, school, or computer laboratory. A LAN allows connected devices to communicate and share resources such as files, printers, and network services.

Key Concepts

- **LAN:** A network covering a limited area.
- **Router:** Connects different networks and forwards traffic between them.
- **Switch:** Connects devices within a LAN and forwards Ethernet frames to the appropriate destination.
- **IP address:** Identifies a device's network interface for IP communication.
- **MAC address:** A link-layer address used to deliver frames within a local network.
- **Ethernet and Wi-Fi:** Common technologies used to connect devices to a LAN.

LAN Communication

When a device communicates with another device on the same IP subnet, it typically determines the destination's link-layer address using ARP in IPv4 networks. The device then sends the data through Ethernet or Wi-Fi. If the destination is outside the local subnet, the device normally sends the traffic to its default gateway.

Security Relevance

LAN security is important because compromised devices may attempt to access other devices or services on the same network. Network defenders can reduce risks by using network segmentation, strong Wi-Fi encryption, access controls, updated devices, and network traffic monitoring.

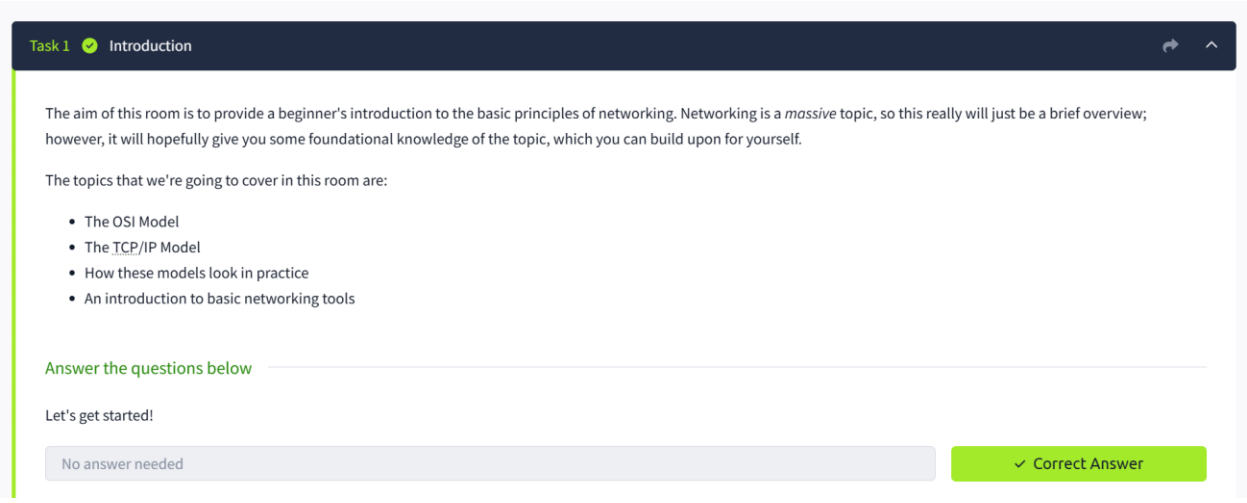
Conclusion

This topic introduces the role of Local Area Networks in connecting devices and enabling communication within a limited area. Understanding LANs, routers, switches, IP addresses, and MAC addresses provides a foundation for learning network security and investigating suspicious traffic.

Room completed: 100%



OSI CONCEPTS



Objective

To understand the basic principles of computer networking, including the OSI model, TCP/IP model, their practical applications, and basic networking tools.

Introduction

Networking is the process of connecting computers and other devices so they can communicate and share information. It plays an important role in cybersecurity because network defenders need to understand how data travels between devices and how network communication can be monitored for suspicious activity.

This TryHackMe room provides a beginner-level introduction to networking concepts and builds a foundation for further learning.

Topics Covered

1. OSI Model

The Open Systems Interconnection (OSI) model is a conceptual framework that explains network communication through seven layers. Each layer has a specific role in transmitting and receiving data.

The seven layers are:

1. Physical
2. Data Link
3. Network
4. Transport
5. Session
6. Presentation
7. Application

The OSI model helps network professionals understand communication processes and troubleshoot network problems systematically.

2. TCP/IP Model

The TCP/IP model describes the protocols and processes used for communication over modern networks, including the Internet. It is commonly represented using four layers:

1. Network Access: Handles communication over the local network, including Ethernet and Wi-Fi.

2. Internet: Handles logical addressing and routing using IP.
3. Transport: Provides communication between applications using protocols such as TCP and UDP.
4. Application: Supports network services and protocols such as HTTP, HTTPS, and DNS.

3. Models in Practice

The OSI and TCP/IP models help explain how data travels from one device to another. For example, when a user visits a website, the browser uses an application protocol such as HTTPS. The transport layer commonly uses TCP, IP routes packets to the destination, and the network access layer sends data across the local network.

These layers work together to deliver information to the correct destination.

4. Basic Networking Tools

Networking tools help users test connectivity, identify devices, examine network services, and troubleshoot communication problems. Examples include:

- Ping: Tests connectivity using ICMP.
- Traceroute/Tracert: Shows the network path toward a destination.
- IP configuration tools: Display local IP and network settings.
- Nmap: Discovers hosts and identifies open ports on authorized targets.
- Wireshark: Captures and analyzes network packets.

These tools are useful for both network troubleshooting and cybersecurity monitoring.

Cybersecurity Relevance

Understanding networking models helps security analysts interpret network traffic and identify unusual communication. Knowledge of IP addresses, protocols, ports, and diagnostic tools supports tasks such as network monitoring, incident investigation, and troubleshooting.

Learning Outcomes

After studying this room, I gained an introductory understanding of the OSI and TCP/IP models, how networking layers work together, and how basic tools can be used to investigate network connectivity. These concepts provide a foundation for further practical work in networking and cybersecurity.

Conclusion

This room introduced the fundamental principles of computer networking and explained the purpose of the OSI and TCP/IP models. I learned how network communication can be

understood in layers and how basic networking tools assist with connectivity testing and troubleshooting. This knowledge will support my future cybersecurity learning, particularly in network traffic analysis and defensive monitoring.

Task 2 The OSI Model: An Overview

The OSI (**O**pen **S**ystems **I**nterconnection) Model is a standardised model which we use to demonstrate the theory behind computer networking. In practice, it's actually the more compact TCP/IP model that real-world networking is based off; however the OSI model, in many ways, is easier to get an initial understanding from.

The OSI model consists of seven layers:

OSI:
Application
Presentation

Answer the questions below

Which layer would choose to send data over TCP or UDP? Answer with the number of the layer: e.g. if the answer would be "the application layer", then you would enter "7".

4 ✓ Correct Answer

Which layer checks received information to make sure that it hasn't been corrupted? Answer with the number of the layer: e.g. if the answer would be "the application layer", then you would enter "7".

2 ✓ Correct Answer

In which layer would data be formatted in preparation for transmission? Answer with the number of the layer: e.g. if the answer would be "the application layer", then you would enter "7".

2 ✓ Correct Answer

Which layer transmits and receives data? Answer with the number of the layer: e.g. if the answer would be "the application layer", then you would enter "7".

1 ✓ Correct Answer

Which layer encrypts, compresses, or otherwise transforms the initial data to give it a standardised format? Answer with the number of the layer: e.g. if the answer would be "the application layer", then you would enter "7".

6 ✓ Correct Answer

Which layer tracks communications between the host and receiving computers? Answer with the number of the layer: e.g. if the answer would be "the application layer", then you would enter "7".

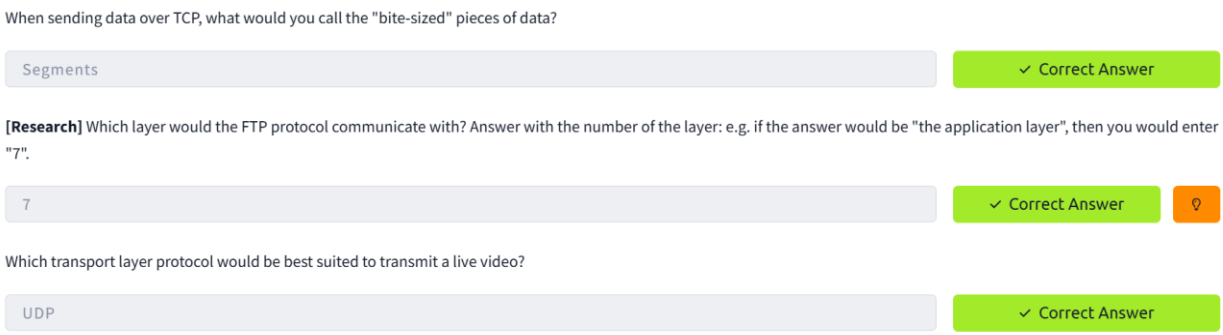
5 ✓ Correct Answer

Which layer accepts communication requests from applications? Answer with the number of the layer: e.g. if the answer would be "the application layer", then you would enter "7".

7 ✓ Correct Answer

Which layer handles logical addressing? Answer with the number of the layer: e.g. if the answer would be "the application layer", then you would enter "7".

3 ✓ Correct Answer



Objective: To understand the seven layers of the OSI model and the role each layer plays in network communication.

Introduction: The Open Systems Interconnection (OSI) model is a seven-layer conceptual framework used to understand how data travels between devices across a network. Each layer performs specific functions and interacts with the layers above and below it.

Layer	Name	Main Function	Examples
7	Application	Provides network services to applications	HTTP, HTTPS, FTP, DNS
6	Presentation	Handles data formatting, encryption, and compression	Data encoding, TLS-related presentation concepts
5	Session	Establishes and manages communication sessions	Session management
4	Transport	Provides end-to-end transport using TCP or UDP	TCP, UDP
3	Network	Handles logical addressing and routing	IP, ICMP
2	Data Link	Handles local delivery, MAC addressing, and frame integrity checks	Ethernet, Wi-Fi, MAC
1	Physical	Transmits and receives bits as electrical, optical, or radio signals	Cables, radio signals

Important Concepts Learned:

- TCP is connection-oriented and supports reliable delivery. Its data units are called segments.
- UDP is connectionless and is commonly useful for real-time applications such as live video, where low latency is important.
- The Network layer handles logical addressing through IP addresses.

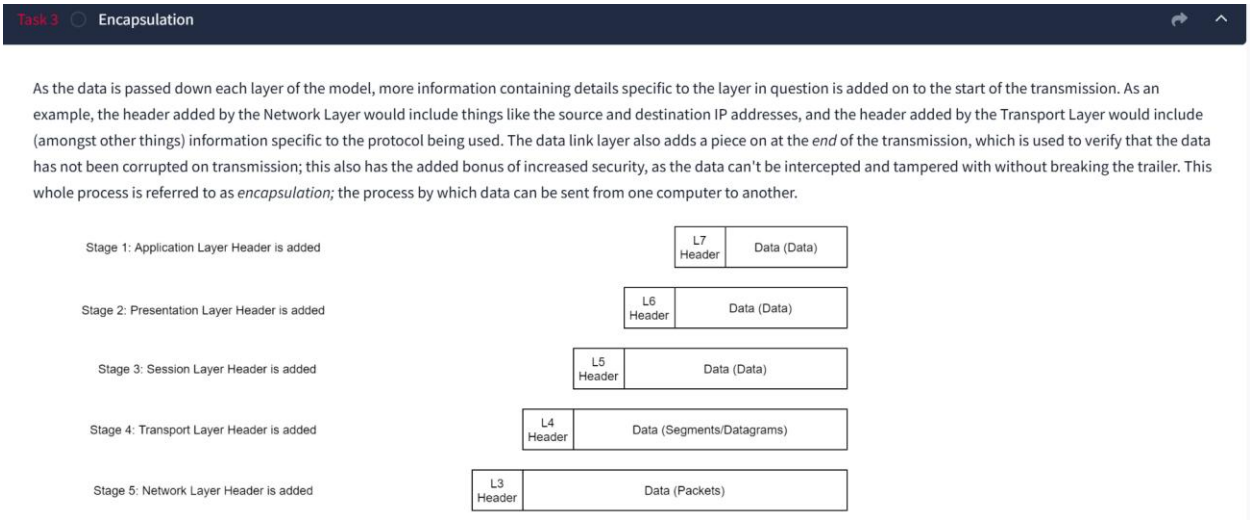
- The Data Link layer handles local network delivery using MAC addresses.
- The Physical layer carries bits through the transmission medium.
- FTP is an Application-layer protocol.

Cybersecurity Relevance:

Understanding the OSI model helps security analysts identify where a network issue or security event may occur. For example, suspicious IP traffic can be investigated at the Network layer, while unusual application requests may need analysis at the Application layer.

Conclusion:

This task helped me understand the functions of all seven OSI layers and how they work together to support communication. I also learned the differences between TCP and UDP, the role of IP and MAC addresses, and how the OSI model helps with network troubleshooting and security analysis.



Answer the questions below

How would you refer to data at layer 2 of the encapsulation process (with the OSI model)?

Frames ✓ Correct Answer

How would you refer to data at layer 4 of the encapsulation process (with the OSI model), if the UDP protocol has been selected?

Datagrams ✓ Correct Answer

What process would a computer perform on a received message?

De-encapsulation ✓ Correct Answer

Which is the only layer of the OSI model to add a *trailer* during encapsulation?

Data Link ✓ Correct Answer

Does encapsulation provide an extra layer of security **(Aye/Nay)**?

Aye ✓ Correct Answer

Objective

To understand how data is prepared for transmission across a network and how the receiving device processes it using the OSI model.

Introduction

Encapsulation is the process of adding protocol-related information to data as it moves down the OSI layers before transmission. Each layer adds information required for communication. At the receiving end, the reverse process, called de-encapsulation, removes this information as data moves up the layers.

Encapsulation Process

During encapsulation, data passes through different layers of the OSI model:

- **Application, Presentation and Session Layers (Layers 7–5):** These layers prepare and manage the application data.
- **Transport Layer (Layer 4):** TCP divides data into segments, while UDP uses datagrams.
- **Network Layer (Layer 3):** Adds logical addressing information, such as source and destination IP addresses. The data unit is called a packet.
- **Data Link Layer (Layer 2):** Adds frame information, including MAC addresses, and a trailer used for error detection. The data unit is called a frame.
- **Physical Layer (Layer 1):** Transmits the data as bits over a physical or wireless medium.

Task Questions and Answers

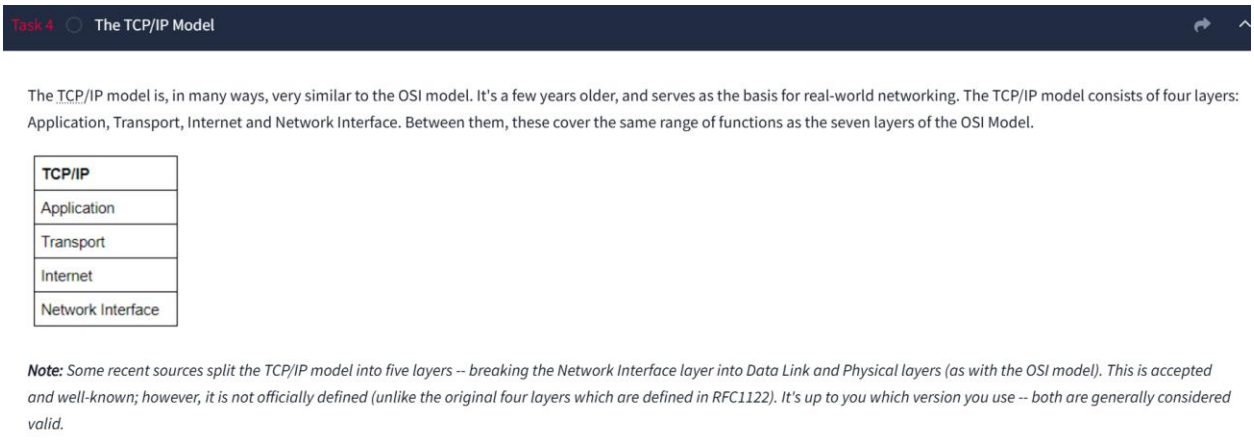
1. **How is data at Layer 2 referred to?**
   Answer: Frame
2. **How is data at Layer 4 referred to when UDP is selected?**
   Answer: Datagram
3. **What process does a computer perform on a received message?**
   Answer: De-encapsulation
4. **Which OSI layer adds a trailer during encapsulation?**
   Answer: Data Link Layer (Layer 2)
5. **Does encapsulation provide an extra layer of security?**
   Answer: Nay

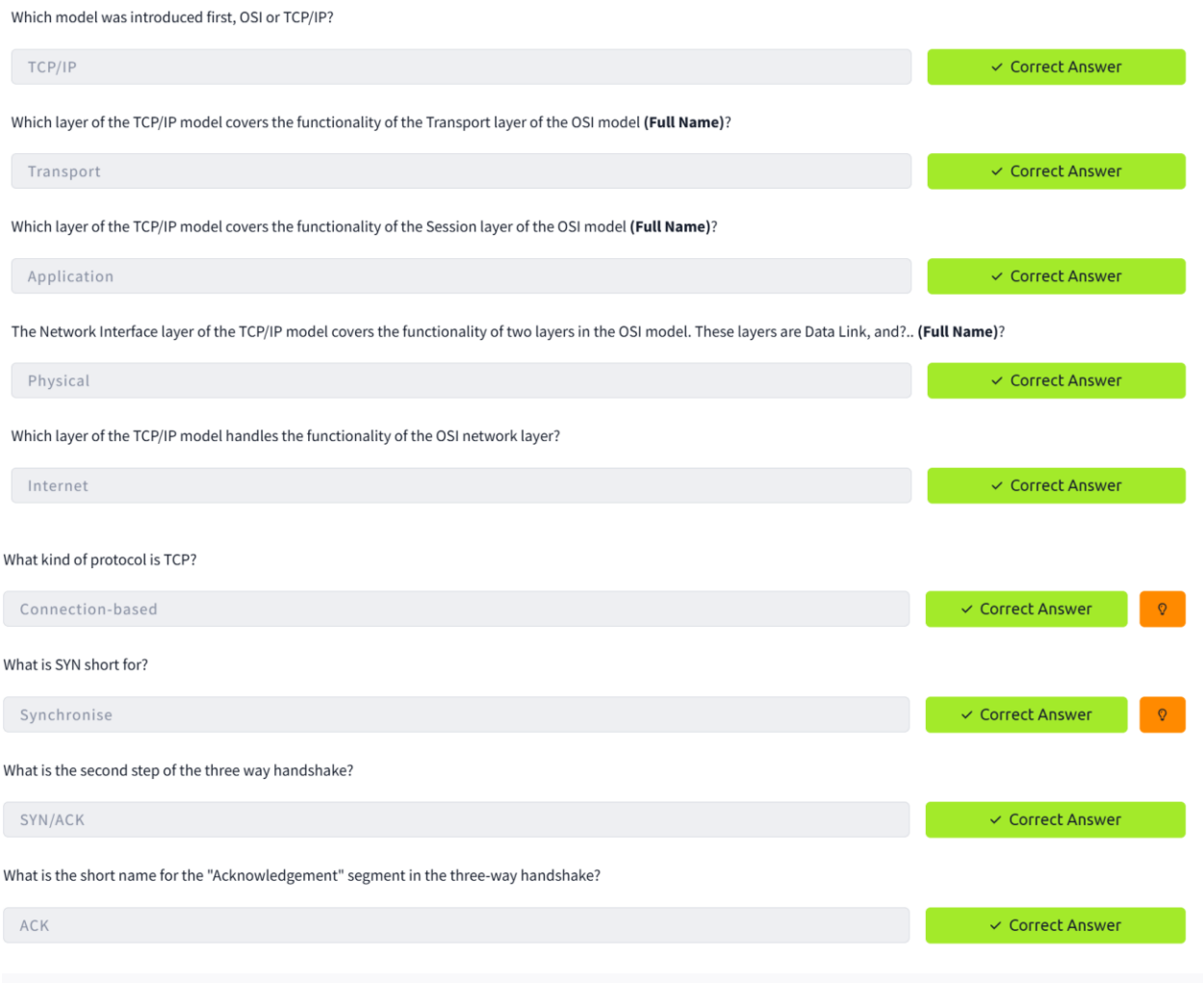
Cybersecurity Relevance

Understanding encapsulation helps security analysts interpret network traffic in tools such as Wireshark. They can inspect Ethernet frames, IP packets, and TCP segments or UDP datagrams to understand how devices communicate. The Data Link Layer trailer commonly supports error detection, but encapsulation alone does not provide encryption or guarantee confidentiality.

Conclusion

This task explained how data is encapsulated as it moves through the OSI layers and de-encapsulated when received. It also clarified the names of data units at different layers and the role of the Data Link Layer in adding a trailer. These concepts are useful for network troubleshooting and cybersecurity traffic analysis.





Objective

To understand the TCP/IP model, its relationship with the OSI model, and how TCP establishes a reliable connection using the three-way handshake.

Introduction

The TCP/IP model is a networking framework used to describe how data travels between devices over the Internet. It was developed before the OSI model and contains four layers: Application, Transport, Internet, and Network Interface. Each layer performs specific functions to support network communication.

TCP/IP Model Layers

TCP/IP Layer	Main Function	Related OSI Layer(s)
Application	Provides network services to applications, such as HTTP, HTTPS, DNS, and FTP	Application, Presentation, Session
Transport	Provides communication between hosts using TCP or UDP	Transport
Internet	Handles logical addressing and routing using IP	Network
Network Interface	Handles local network delivery and physical transmission	Data Link, Physical

TCP and UDP

Transmission Control Protocol (TCP) is a connection-oriented transport protocol. It establishes a connection before transferring data and provides reliable, ordered delivery through acknowledgements and retransmission.

User Datagram Protocol (UDP) is connectionless and has lower communication overhead. It is often used when low latency is important, such as live video and voice communication.

TCP Three-Way Handshake

TCP uses a three-way handshake to establish a connection between a client and a server.

1. **SYN (Synchronize):** The client sends a SYN segment to request a connection.
2. **SYN-ACK (Synchronize-Acknowledge):** The server responds with SYN-ACK to acknowledge the request and synchronize its sequence number.
3. **ACK (Acknowledgement):** The client sends an ACK to confirm the response. The connection is then established.

Handshake sequence: Client → SYN → Server; Server → SYN-ACK → Client; Client → ACK → Server.

Task Questions and Answers

1. **Which model was introduced first?**

   TCP/IP.

2. **Which TCP/IP layer covers the OSI Transport layer?**

   Transport.

3. **Which TCP/IP layer covers the OSI Session layer?**

   Application.

4. **Which OSI layer, along with Data Link, is covered by Network Interface?**

   Physical.

5. **Which TCP/IP layer handles OSI Network layer functionality?**

   Internet.

6. **What kind of protocol is TCP?**

   Connection-oriented.

7. **What is SYN short for?**

   Synchronize.

8. **What is the second step of the three-way handshake?**

   SYN-ACK.

9. **What is the short name for the Acknowledgement segment?**

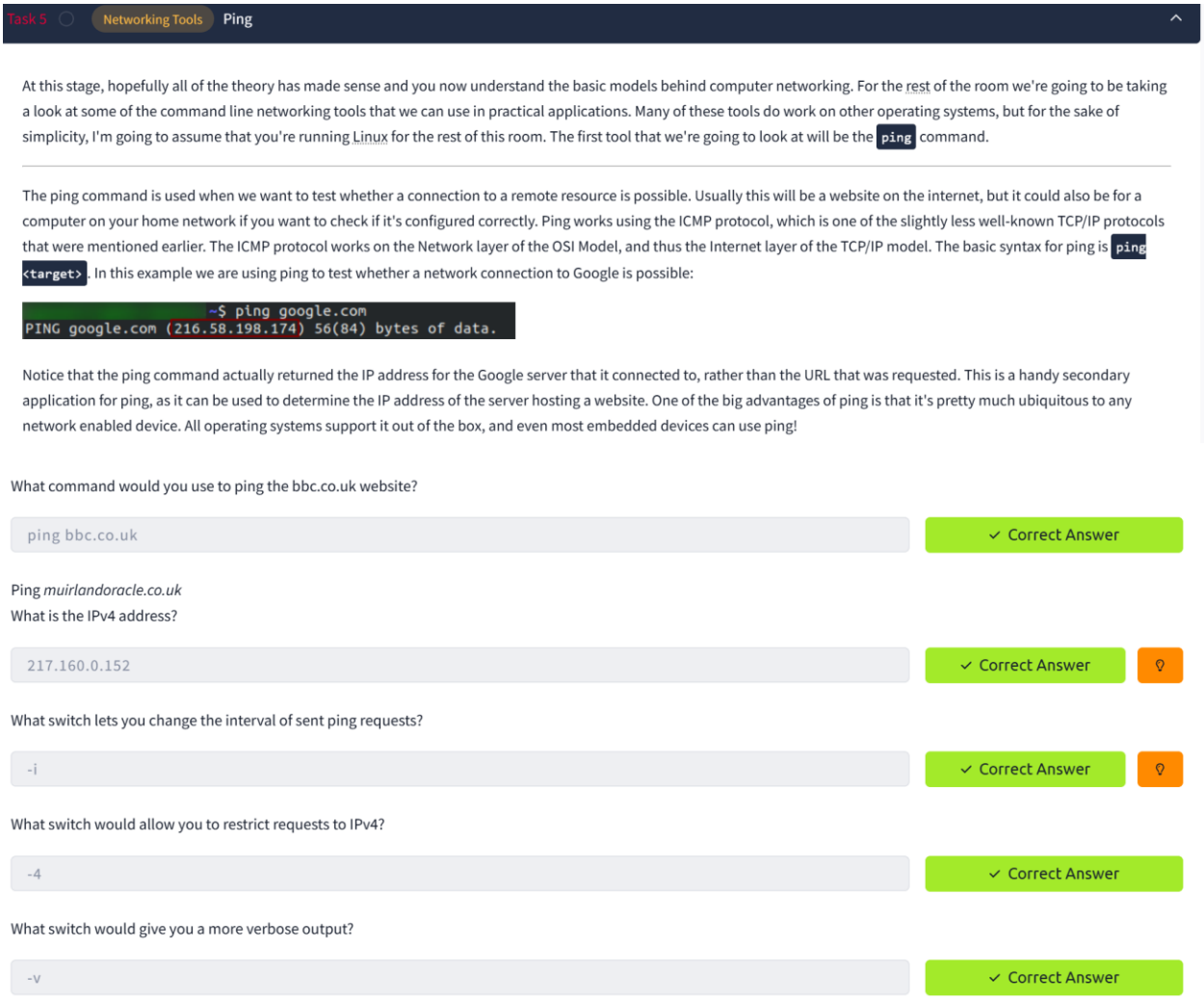
   ACK.

Cybersecurity Relevance

Understanding TCP/IP and the three-way handshake helps security analysts interpret network traffic in packet-analysis tools such as Wireshark. Repeated SYN requests without completed handshakes may indicate a SYN flood attempt, although similar patterns can also occur for legitimate reasons. Analysts should investigate the source, frequency, and surrounding traffic before deciding whether an alert is malicious.

Conclusion

This task explained the four layers of the TCP/IP model, the differences between TCP and UDP, and the TCP three-way handshake. These concepts provide a foundation for understanding packet captures, troubleshooting network connections, and identifying potentially suspicious traffic.



Objective

To understand the purpose of the ping command, the role of ICMP in network troubleshooting, and the basic options used to control ping requests.

Introduction

Ping is a network diagnostic command used to check whether a destination device is reachable over an IP network. It sends ICMP Echo Request messages and listens for ICMP Echo Reply messages. The results can help identify connectivity problems and measure round-trip time.

Practical Activity

During the TryHackMe networking exercise, I studied how the ping command can be used to test connectivity with a website such as bbc.co.uk. A ping command can be written as:

ping bbc.co.uk

The destination hostname is resolved to an IP address, and the system sends ping requests to test reachability. If replies are received, the output usually displays the response time and other network information. If no replies are received, the destination may be unreachable or configured to block ICMP traffic.

The hostname muirlandoracle.co.uk was also used in the exercise to understand IPv4 address resolution. Its IPv4 address can be checked using a DNS lookup tool or a terminal command such as ping -4 muirlandoracle.co.uk. Since DNS records can change, the current resolved address should be verified when possible.

Common Ping Options

The -i option on Linux changes the interval between ping requests. The -4 option forces the use of IPv4. The -v option requests verbose output where supported by the command implementation.

Cybersecurity Relevance

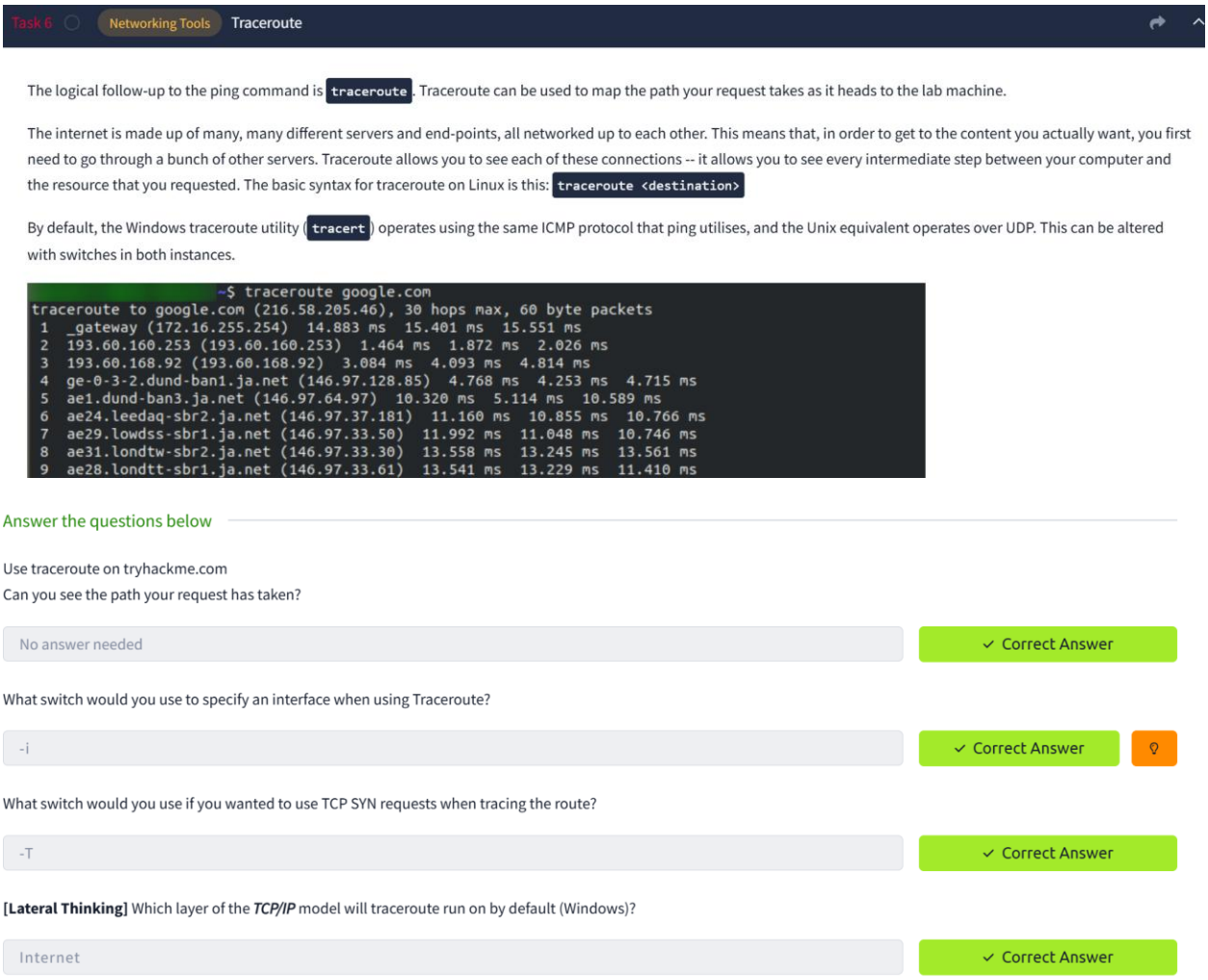
Ping is useful for basic network troubleshooting and checking whether systems respond to network traffic. Security analysts may also use ICMP traffic to understand network activity. However, ping results alone cannot prove that a system is secure or compromised. Some legitimate systems block ICMP requests, while unusual ICMP activity may require further investigation.

Findings

The exercise introduced the purpose of ICMP Echo Requests and Echo Replies, hostname resolution, and the use of command-line options to control ping behaviour. It also demonstrated that network diagnostic results depend on connectivity, DNS resolution, and the destination's security settings.

Conclusion

This task provided an introduction to ping and ICMP as basic network diagnostic tools. Understanding these concepts supports network troubleshooting and helps beginners recognise normal connectivity behaviour before investigating potentially suspicious network activity.



Objective

To understand how traceroute identifies the path packets take from a source device to a destination and how it helps with network troubleshooting.

Introduction

Traceroute is a network diagnostic tool that helps identify the intermediate routers between a source device and a destination. It displays the route hop by hop and can help locate delays or connectivity problems along the path.

On Windows, the command is called tracert. On Linux, it is commonly called traceroute.

Practical Activity

The TryHackMe exercise introduced tracing the route to tryhackme.com. The tool sends probes with increasing Time To Live (TTL) values. When a probe's TTL expires at an intermediate

router, that router may return an ICMP Time Exceeded message. By repeating this process with higher TTL values, traceroute can identify successive hops toward the destination.

A typical Windows command is:

tracert tryhackme.com

On Linux, the corresponding command is:

traceroute tryhackme.com

The output lists the hops and the response times for each probe. Some hops may show timeouts because routers or firewalls do not respond to traceroute probes. Therefore, missing responses do not necessarily mean the network path is broken.

Important Traceroute Options

On Linux, the -i option can specify the network interface to use. The -T option selects TCP SYN probes in implementations that support it. Options may differ between operating systems and traceroute versions.

On Windows, the default tracert command uses ICMP Echo Requests. In the four-layer TCP/IP model, ICMP is associated with the Internet layer.

Cybersecurity Relevance

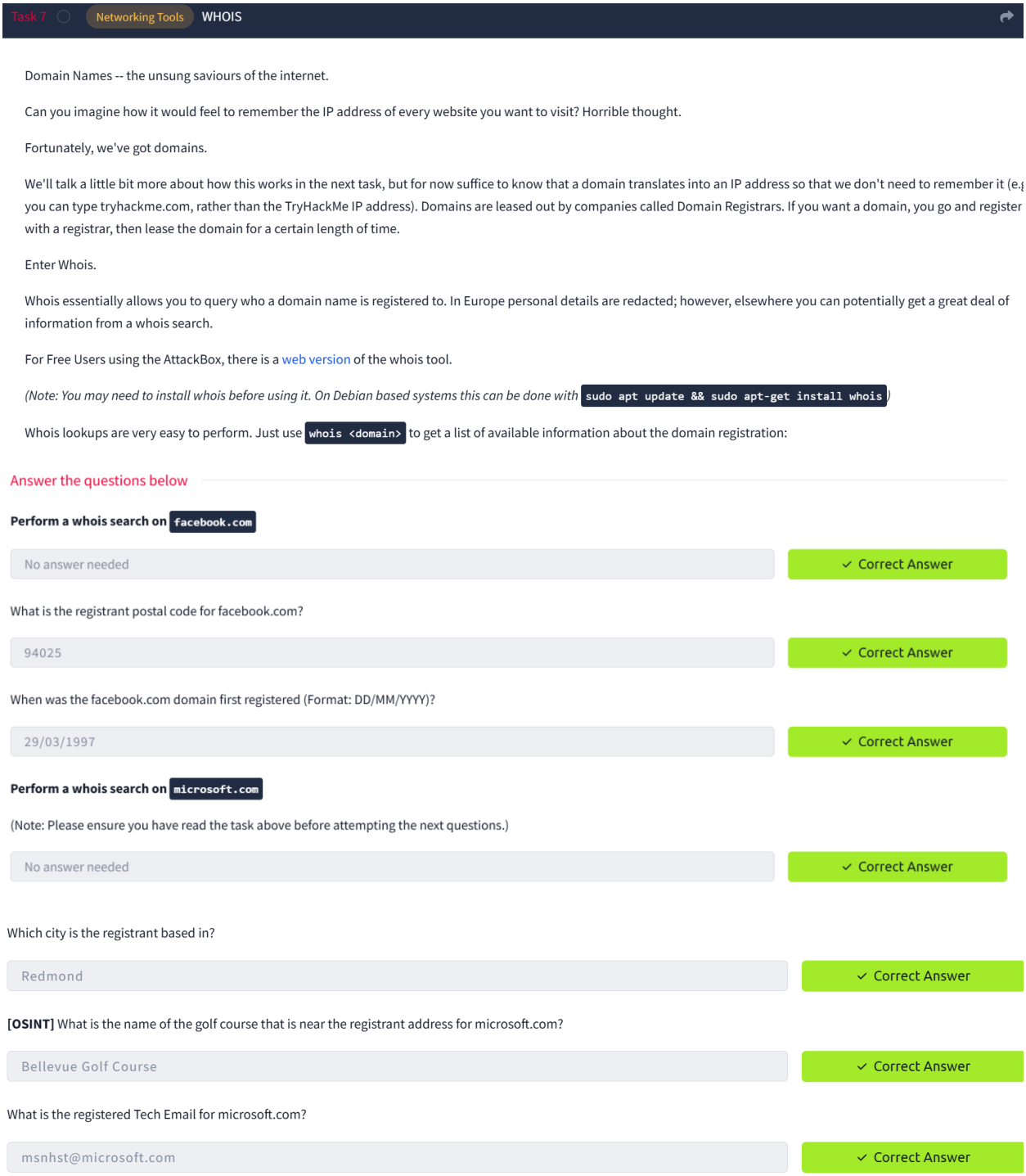
Traceroute helps network defenders understand routing paths and investigate connectivity problems. Unexpected route changes or unusual network paths may deserve further investigation, but traceroute alone cannot confirm malicious activity. Results should be interpreted alongside DNS information, firewall logs, routing data, and other network evidence.

Findings

This task introduced hop-by-hop route discovery, the role of TTL, and the use of traceroute for diagnosing network connectivity. It also highlighted that the probes and available command-line options depend on the operating system and tool implementation.

Conclusion

Traceroute is a useful network diagnostic tool for examining the path between a source and destination. Understanding its operation helps beginners troubleshoot connectivity issues and interpret network behaviour from a defender's perspective.



Objective

To understand WHOIS lookups and learn how to collect publicly available domain registration information for basic network reconnaissance.

Introduction

WHOIS is a query tool used to retrieve domain registration information. Depending on the domain and privacy settings, the results may include the registrar, registration and expiry dates, registrant details, name servers, and technical contact information.

WHOIS is commonly used during the reconnaissance phase of cybersecurity assessments to understand a domain's public registration footprint.

Tools Used

- TryHackMe — Introductory Networking room
- WHOIS lookup service
- Public search engine or map service for the OSINT activity

Practical Activity

1. WHOIS Lookup for facebook.com

I reviewed the publicly available registration information for facebook.com. The exercise focused on identifying the registrant postal code and the original domain registration date.

The postal code provided in the exercise was **94025**, and the domain's first registration date was **29/03/1997**.

This activity demonstrated how WHOIS information can help analysts identify domain registration details and understand the historical background of a domain.

2. WHOIS Lookup for microsoft.com

Next, I examined the WHOIS information for microsoft.com. The exercise required identifying the registrant's city and the technical contact email address.

The city listed in the exercise was **Redmond**. The registered technical contact email was **msnhst@microsoft.com**.

These details demonstrated how domain registration records may expose organisational information that can support reconnaissance and security research.

3. Open-Source Intelligence (OSINT)

The task also involved identifying a golf course near the registrant address associated with microsoft.com. Using publicly available location information and a map or search service, the exercise identified **Bellevue Golf Course**.

This activity introduced OSINT, which involves collecting and analysing information from publicly accessible sources. OSINT findings should be checked against reliable sources because addresses and registration records can change.

Cybersecurity Relevance

WHOIS and OSINT are useful during authorised security assessments because they help analysts understand a domain's registration history, registrar, technical contacts, and publicly visible organisational information.

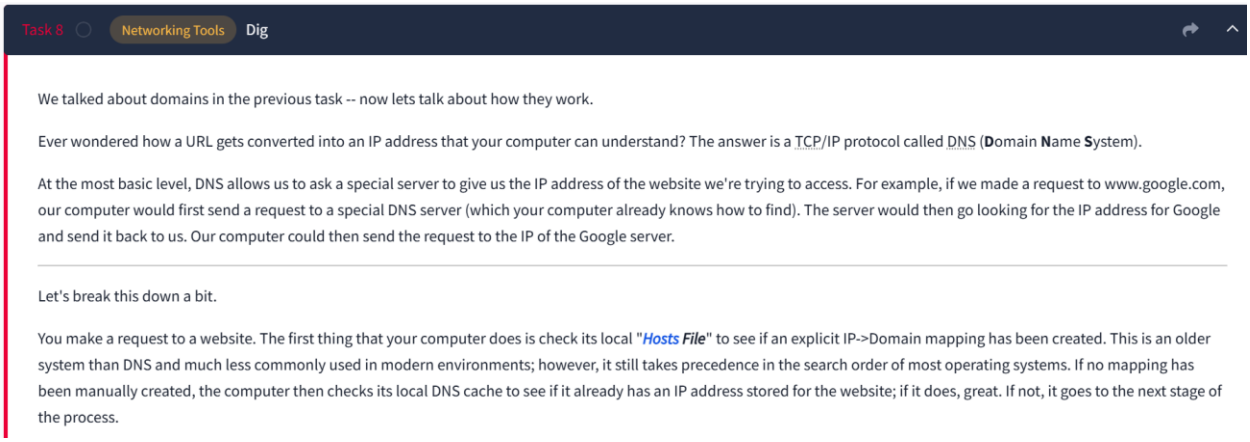
However, registration information may be outdated, privacy-protected, or different across lookup services. Analysts should verify findings before using them in a security report. Public information should be handled responsibly and only used within an authorised assessment.

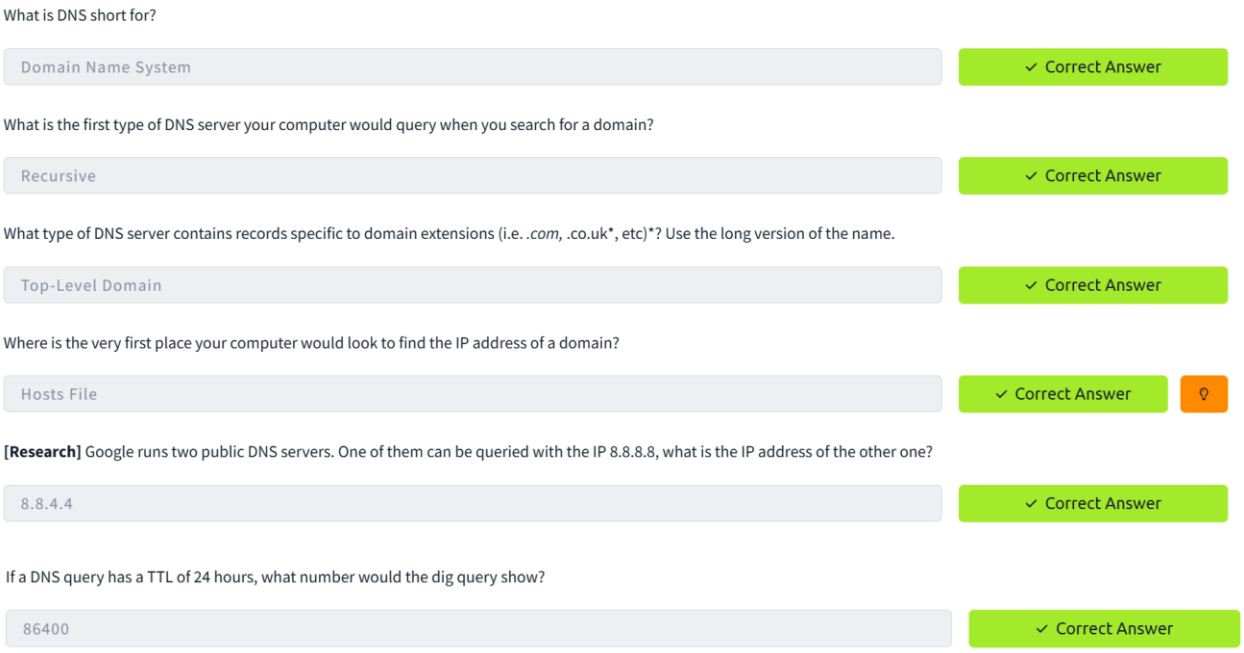
Findings

The exercise demonstrated how WHOIS lookups can reveal domain registration details and how OSINT techniques can connect public registration information with geographic information. It also highlighted the importance of verifying information and understanding the limitations of public records.

Conclusion

This task introduced WHOIS as a basic reconnaissance tool and demonstrated how it can be used alongside OSINT techniques. These skills provide a foundation for gathering domain intelligence, documenting findings, and supporting defensive cybersecurity investigations.





Objective

To understand the Domain Name System (DNS), learn how domain names are resolved into IP addresses, and explore the purpose of the dig command.

Introduction

The Domain Name System (DNS) translates human-readable domain names, such as tryhackme.com, into IP addresses that computers use to communicate over networks. Without DNS, users would generally need to remember the IP addresses of websites.

The dig command is a DNS lookup utility commonly available on Linux. It helps users query DNS records and inspect information such as IP addresses, record types, name servers, and Time to Live (TTL) values.

DNS Resolution Process

When a user requests a domain name, the computer first checks its local hosts file and other local resolution mechanisms. If the address is not already available, the computer typically sends a query to a recursive DNS resolver.

If the resolver does not have a valid cached answer, it may query a root DNS server, a Top-Level Domain (TLD) server, and the domain's authoritative DNS server. The authoritative server provides the relevant DNS record, and the resolver returns the answer to the computer.

The result may then be cached for a period determined by the record's TTL.

Practical Activity

During the TryHackMe exercise, I studied the purpose of DNS and the role of different DNS servers in domain resolution. The activity covered recursive resolvers, TLD servers, hosts files, public DNS services, and TTL values.

The following command can be used on a Linux terminal to query a domain:

dig tryhackme.com

To request an IPv4 address record, the following command can be used:

dig tryhackme.com A

The output may include the DNS question, answer, name servers, query time, and TTL information. A TTL of 24 hours is equivalent to **86,400 seconds**.

Google provides public DNS resolvers at 8.8.8.8 and 8.8.4.4.

Cybersecurity Relevance

DNS analysis is useful in cybersecurity because it helps analysts investigate domain resolution, identify unexpected DNS responses, and examine potentially suspicious domain names. DNS logs can also help identify repeated queries to unusual domains or possible command-and-control activity.

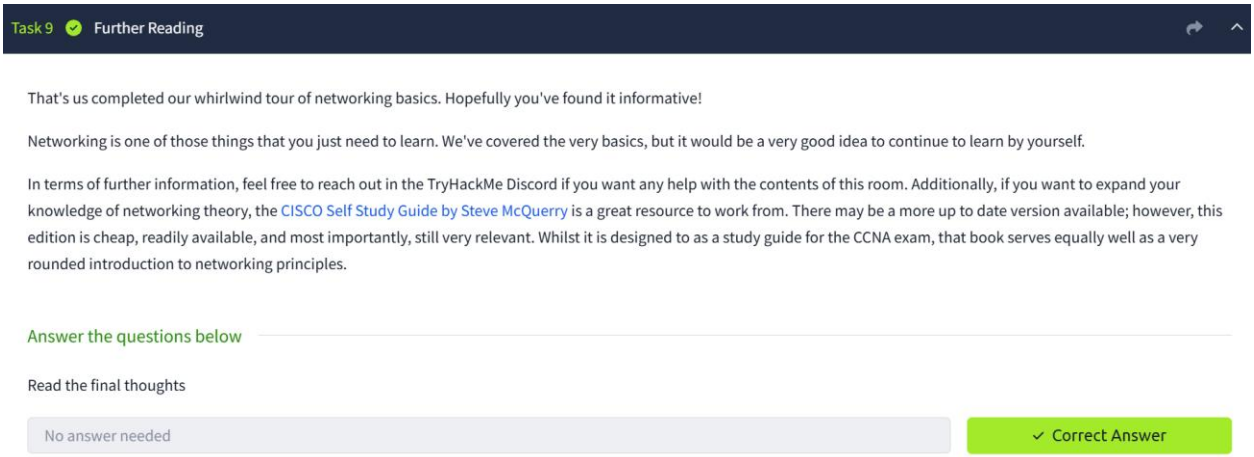
However, an unusual domain or DNS response is not automatically malicious. Analysts should examine context, reputation, query patterns, and other available evidence before drawing conclusions.

Findings

This task demonstrated how DNS resolves domain names, how recursive and authoritative DNS services work together, and how TTL values control the caching period of DNS records. It also introduced dig as a useful command for inspecting DNS information.

Conclusion

The exercise provided a foundation for understanding DNS resolution and basic DNS investigation. These concepts are useful for troubleshooting connectivity problems and analysing DNS traffic from a cybersecurity defender's perspective.



Objective

To review the networking concepts covered in the TryHackMe Introductory Networking room and identify resources for continued learning.

Overview

The Further Reading section summarised the importance of continuing to develop networking knowledge. It recommended using additional learning resources and the TryHackMe community to improve understanding of networking principles.

The lesson highlighted that networking fundamentals require regular study and practice. Understanding how devices communicate, how protocols operate, and how data travels across networks provides a strong foundation for more advanced cybersecurity topics.

Key Learning Outcomes

After completing the room, I reviewed the following concepts:

- Network devices, IP addresses, and MAC addresses.
- Ping and ICMP for basic connectivity testing.
- The seven layers of the OSI model and the four layers of the TCP/IP model.
- Encapsulation and de-encapsulation.
- TCP, UDP, and the TCP three-way handshake.
- Traceroute for examining network paths.
- WHOIS for reviewing domain registration information.
- DNS resolution and the use of dig to inspect DNS records.

Continued Learning

Further networking knowledge can be developed through additional practical labs, documentation, and structured study resources. The lesson also recommended the TryHackMe Discord community for learning support and introduced the Cisco self-study guide by Steve McQuerry as a resource for networking theory.

Conclusion

The Further Reading section reinforced the importance of continuous learning in networking. Completing the room provided a beginner-level introduction to essential networking concepts that support future study in network security, traffic analysis, and security operations.

